Finite-time blowup for a Schrödinger envelope coupled to ionization

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Abstract

We construct a finite-time singularity for a three-dimensional focusing Schrödinger envelope coupled to an ionization density. For one strictly positive ionization coefficient, with cubic absorption and no avalanche term, a nonzero compactly supported smooth optical datum and zero initial density produce a smooth solution whose gradient norm tends to infinity at a finite time. The density remains nonnegative. Thus global smooth propagation for all admissible coefficients and Schwartz data is false. The construction continues a scalar self-similar profile to positive ionization, then solves the optical and density perturbations together. A fixed scalar linear theory and compactly supported finite-dimensional corrections allow the singular solution to start from compact smooth data.

1 Introduction

Ionization modifies a focusing nonlinear Schrödinger envelope through both absorption and a density-dependent defocusing term. Such terms arise in models of laser filamentation; their derivation and the question of global regularization are discussed by Dumas, Lannes and Szeftel [1, Section 4.1.2, equation (54) and Open Problem 3]. We consider the following precise system on $\mathbb{R}^3$, with spatial coordinate $x = (x_1, x_2, z)$:

$$\begin{aligned} i(\partial_t + c_g\partial_z)u + \epsilon\Delta u + \epsilon(|u|^2 - \rho)u &= -i\epsilon c\big(\alpha_4|u|^{2K-2}u + \alpha_5\rho u\big), \\ \partial_t\rho &= \epsilon\alpha_4|u|^{2K} + \epsilon\alpha_5\rho|u|^2. \end{aligned} \tag{1.1}$$

Here u is complex, ρ is real, the Laplacian is elliptic in all three spatial coordinates, and

$$\epsilon, c, c_g, \alpha_4 > 0, \qquad \alpha_5 \geq 0, \qquad K \in \mathbb{Z}, \quad K \geq 2.$$

The universal global-propagation question asks whether every Schwartz pair (u_0, ρ_0) with $\rho_0 \geq 0$ has a solution in $C([0,\infty); H^m \times H^m)$ for every integer $m \geq 3$. This is the formulation recorded as AIM Problem 441.[1] The theorem below gives an admissible counterexample.

For the scalar focusing cubic NLS in three dimensions, Donninger and Schörkhuber [2] construct a self-similar blowup profile by a computer-assisted proof with exact fraction arithmetic. That profile has finite energy but a spatial tail of order r^{-1}, hence is not in L^2. Li [3] develops linear dispersive estimates and finite codimensional stability theory for a class of mass-supercritical self-similar NLS profiles. We use the quantitative profile inverse from the former work and the fixed-profile linear estimates and compactly supported correction modes from the latter. Neither result directly treats the ionization coupling in (1.1).

[1] The precise problem statement is available at the immutable AIM 441 statement.

Theorem 1.1. *There exist $\delta > 0$, a nonzero $u_0 \in C_c^\infty(\mathbb{R}^3;\mathbb{C})$, and $T \in (0,\infty)$ such that, for*

$$\epsilon = c = c_g = 1, \qquad \alpha_4 = \delta, \qquad K = 2, \qquad \alpha_5 = 0, \qquad \rho_0 = 0,$$

the original smooth solution of (1.1) exists on $[0,T)$, belongs to $C_{\mathrm{loc}}([0,T);H^m(\mathbb{R}^3;\mathbb{C})\times H^m(\mathbb{R}^3;\mathbb{R}))$ for every integer $m \geq 3$, has $\rho(t,x) \geq 0$, and satisfies

$$\lim_{t\uparrow T} \|\nabla u(t)\|_{L^2(\mathbb{R}^3)} = \infty.$$

In particular it has no global continuation in $C_t(H^3 \times H^3)$.

One strictly positive coefficient and one Schwartz datum suffice to negate the universal assertion. The theorem selects a finite member of a sequence of small positive coefficients; it gives neither an explicit full Cauchy threshold nor failure for every coefficient. It also makes no open-set stability claim.

There are two obstacles to transferring the scalar construction. First, the density is of the same similarity order as $|u|^2$, so it must be included in the stationary profile. Its radial equation determines a nonlocal integral of the fourth power of the optical profile. We bound this added forcing in the exact spaces of the scalar inverse and preserve the opposite signs of its real obstruction. A compactness argument then selects one scalar limiting profile before any spectral constants are chosen.

Second, a compact cutoff of the profile must be compatible with the unstable finite coordinates, and the evolving density cannot be replaced by its stationary value. We use modified projections whose finite ranges consist of compactly supported smooth functions to make the optical datum compact, and solve the two perturbation equations jointly. Moving with the optical envelope introduces a translation in the density equation; its characteristic propagator retains that term exactly. A mixed space-time estimate controls the quartic density source, and weighting the density perturbation by the inverse square root of the ionization coefficient makes the feedback contractive. The resulting solution approaches a nonzero profile in a shrinking spatial scale, forcing its gradient norm to diverge.

Section 2 constructs the profiles and verifies the hypotheses of the scalar linear theory. Section 3 establishes the linear, product and transport estimates. Section 4 constructs the compact-data solution and identifies it with the smooth physical Cauchy solution. Appendix A describes the finite exact calculation. We write $\mathcal{F}f(\xi) = (2\pi)^{-3/2}\int e^{-ix\cdot\xi}f(x)\,dx$. For $0 < s < 3/2$, $\dot{H}^s$ has norm $\||\xi|^s\mathcal{F}f\|_2$ and its canonical realization in $L^{6/(3-2s)}$; H^s uses the inhomogeneous Fourier weight. All intersection norms are sums, and all spatial norms use Lebesgue measure on $\mathbb{R}^3$.

2 Profiles with positive ionization

We construct profiles for the parameter choice in Theorem 1.1. For a function of the similarity variable $y \in \mathbb{R}^3$, write $\Lambda_a f = af + y\cdot\nabla f$. The stationary equations to be solved first are

$$\Delta Q - Q + i\beta\Lambda_1 Q + (1+i\delta)|Q|^2Q - RQ = 0, \qquad \beta\Lambda_2 R = \delta|Q|^4. \tag{2.1}$$

The density profile is determined by the optical profile. For a radial $Q(y) = q(|y|)$, its regular solution is

$$R(r) = \frac{\delta}{\beta r^2}\int_0^r s|q(s)|^4\,ds. \tag{2.2}$$

This identity makes the added coupling small when δ is small, while keeping δ strictly positive.

2.1 The quantitative scalar construction

We use the concrete scalar construction of Donninger and Schörkhuber [2], with precisely the parameter and polynomial used there. Set

$$\beta_* = \frac{772201763088846}{841768781900003}, \qquad J = [-10^{-10}, 10^{-10}], \qquad r_* = 1.2 \cdot 10^{-6}, \qquad \beta = \beta_* + \theta \quad (\theta \in J).$$

The radial change of variables is

$$r = \frac{1+y}{1-y}, \qquad q(r) = (1+r)^{-1-i/\beta} F(y), \qquad -1 < y < 1. \tag{2.3}$$

In this subsection y is the scalar compactified radial coordinate. The source spaces, denoted here by X_p, Y_p, have norms

$$\|f\|_{X_p} = \|f\|_\infty + \sup_{-1<y<1} (1-y^2)|f'(y)|, \qquad \|g\|_{Y_p} = \sup_{-1<y<1} (1+y)(1-y)^2 |g(y)|.$$

Functions in X_p are continuous on $[-1,1]$, are C^1 in its interior, and have continuously extended weighted derivative. The space Y_p is the source forcing space, which permits the fundamental-matrix join at $y = 0$. Continuous forcings with continuous weighted endpoint limits belong to it.

Here is the precise part of the scalar construction that we need. In the notation of [2], its a_*, a become β_*, θ. If $P_*(y) = \sum_{n=0}^{50} c_n T_n(y)$ is the rational polynomial in its Appendix A, put

$$f_{*,\theta}(y) = P_*(y) - P_*(1) + \frac{2i\beta}{1-i\beta} P_*'(1). \tag{2.4}$$

There are a real-linear map $J_\theta : Y_p \to X_p$, a real functional $\psi_\theta : Y_p \to \mathbb{R}$, an approximate differential operator $\mathcal{L}_\theta$ induced by the source's chosen fundamental matrix, and a nonlinear map G_θ with

$$G_\theta(f) = \mathcal{L}_\theta f - \mathcal{R}_\beta(f_{*,\theta} + f). \tag{2.5}$$

Here $\mathcal{R}_\beta$ is the compactified residual of the scalar equation obtained from (2.1) with $\delta = R = 0$. The second derivatives cancel in (2.5), so this identity is meaningful for $f \in X_p$ before C^2 regularity is known. The inverse property is the following: if $f = J_\theta g$ and $\psi_\theta g = 0$, then $\mathcal{L}_\theta f = g$ away from the artificial join, and the source construction gives C^1 matching across it. The scalar obstruction is exactly the term removed in defining the regularized inverse.

The maps are strongly continuous in θ on fixed inputs. On the closed X_p ball of radius r_*, the quantitative estimates are

$$\|J_\theta\|_{Y_p \to X_p} \le 232, \qquad \|\psi_\theta\|_{Y_p \to \mathbb{R}} \le 13, \tag{2.6}$$

$$\|G_\theta(f)\|_{Y_p} \le 5 \cdot 10^{-6} \|f\| + 10 \|f\|^2 + 3 \|f\|^3 + 5 \cdot 10^{-9}, \tag{2.7}$$

$$\|G_\theta(f) - G_\theta(g)\|_{Y_p} \le \left[5 \cdot 10^{-6} + 10(\|f\| + \|g\|) + 3(\|f\|^2 + \|g\|^2)\right] \|f - g\|. \tag{2.8}$$

Unsubscripted correction norms here are X_p norms. At $\theta_- = -10^{-10}$ and $\theta_+ = 10^{-10}$, respectively,

$$\psi_{\theta_-} \mathcal{R}_{\beta_* + \theta_-}(f_{*,\theta_-}) \ge 4 \cdot 10^{-10}, \qquad \psi_{\theta_+} \mathcal{R}_{\beta_* + \theta_+}(f_{*,\theta_+}) \le -4 \cdot 10^{-10}. \tag{2.9}$$

These are results for the scalar operator, rather than a coupled profile theorem. The inverse and continuity statements are Definition 3.1 and Lemmas 3.27, 3.29, 3.35, 3.37–3.39 of [2]; the bounds are Lemma 7.5, Proposition 7.6, Lemma 10.1 and Proposition 10.2. The strict signs are Corollary 9.23. We use the weaker residual allowance displayed in Proposition 10.2; Proposition 6.4 gives $4 \cdot 10^{-9}$.

2.2 Adding the density and absorption terms

For $F = f_{*,\theta} + f$, define q by (2.3) and R_F by (2.2). Since $dy/dr = (1-y)^2/2$, the additional compactified residual is exactly

$$P_{\delta,\theta}(F) = \frac{i\delta F|F|^2}{(1-y)^2} - \frac{4R_F(r(y))F}{(1-y)^4}. \tag{2.10}$$

Indeed, dividing the additional physical terms by $(1+r)^{-1-i/\beta}(dy/dr)^2$ gives this formula. If $M \geq \|F\|_\infty$ and $\beta_{\min} = \beta_* - 10^{-10} > 9/10$, then

$$0 \leq R_F(r) \leq \frac{2\delta M^4}{\beta(1+r)^2}, \tag{2.11}$$

$$\|P_{\delta,\theta}(F)\|_{Y_p} \leq \delta(2M^3 + 4M^5/\beta_{\min}), \tag{2.12}$$

$$\|P_{\delta,\theta}(F) - P_{\delta,\theta}(G)\|_{Y_p} \leq \delta(6M^2 + 20M^4/\beta_{\min}) \|F - G\|_{X_p}. \tag{2.13}$$

To see the density bound, put $I(r) = \int_0^r s(1+s)^{-4}\,ds$. For $r \leq 1$, $I(r) \leq r^2/2$; for $r \geq 1$, $I(r) \leq 1/6$. Thus $(1+r)^2 r^{-2} I(r) \leq 2$. Multiplication by the Y_p weight in (2.10) gives (2.12); for the density term the remaining factor is $4(1+y)/(1-y)^2 = 2r(1+r)$. Factoring the cubic and fourth-power differences gives (2.13). At $r = 0$ the weighted density term tends to zero. At infinity the integral $\int_0^r s|q(s)|^4\,ds$ converges absolutely, so the weighted forcing has a finite endpoint limit. Thus the added residual belongs to Y_p.

The finite rational inequalities proved in Appendix A give, throughout the correction ball,

$$\|F\|_\infty < 50, \qquad \operatorname{Re} F(-1) < -1, \qquad |f_{*,\theta}(1)| > 1/10. \tag{2.14}$$

For $0 < \delta \leq 10^{-23}$, equations (2.12) and (2.13) therefore give the convenient bounds $p = 10^{-13}$ and $l_P = 10^{-6}$ for the forcing and its Lipschitz constant. Define

$$T_{\delta,\theta} f = J_\theta[G_\theta(f) - P_{\delta,\theta}(f_{*,\theta} + f)].$$

The self-map bound is

$$232(5 \cdot 10^{-6} r_* + 10 r_*^2 + 3 r_*^3 + 5 \cdot 10^{-9} + p) < r_*.$$

Its Lipschitz constant is at most

$$232(5 \cdot 10^{-6} + 20 r_* + 6 r_*^2 + l_P) < 1/100. \tag{2.15}$$

The contraction theorem gives a unique f_θ^δ in this ball, continuous in θ and δ, including $\delta = 0$. Here strong continuity of J_θ on fixed inputs suffices: subtract the two fixed-point equations and absorb their common contraction constant.

For $\delta = 0$, equations (2.7)–(2.9) show that $\psi_\theta G_\theta(f_\theta^0)$ has negative and positive endpoint signs, with margin

$$m_0 = 4 \cdot 10^{-10} - 13(5 \cdot 10^{-6} r_* + 10 r_*^2 + 3 r_*^3) > 10^{-10}.$$

Moreover, $\left\| f_\theta^\delta - f_\theta^0 \right\|_{X_p} < 235p$. Consequently the continuous obstruction

$$\Psi_\delta(\theta) = \psi_\theta[G_\theta(f_\theta^\delta) - P_{\delta,\theta}(f_{*,\theta} + f_\theta^\delta)]$$

changes at either endpoint by at most

$$13\left[1+\frac{232(5\cdot 10^{-6}+20r_*+6r_*^2)}{1-1/100}\right]p<10^{-11}.$$

Its signs persist. The intermediate value theorem gives a root $\theta_\delta\in J$ for each such positive δ.

At the root the regularization term vanishes. The inverse property and (2.5) imply $\mathcal{R}_\beta(F)+P_{\delta,\theta}(F)=0$ on the two sides of the join. The correction is C^1 there. After cancellation of the artificial coefficients, its physical second-order ODE has continuous forcing across the join, and hence gives a C^2 solution. We obtain

$$q''+(2/r+i\beta r)q'+(i\beta-1)q+(1+i\delta)|q|^2q-R_Fq=0. \tag{2.16}$$

This establishes the coupled profile by the scalar inverse and a new forcing estimate; no differentiability of the selected parameter root is needed.

2.3 Regularity and a fixed scalar limit

The source norm implies uniformly $|q(r)|\le C(1+r)^{-1}$ and $|q'(r)|\le C/[r(1+r)]$. At the origin put $A(q,R)=-(i\beta-1)q-(1+i\delta)|q|^2q+Rq$. Integrating (2.16) gives

$$q'(r)=r\int_0^1\tau^2e^{i\beta r^2(\tau^2-1)/2}A(q(r\tau),R(r\tau))\,d\tau, \tag{2.17}$$

$$R(x)=\frac{\delta}{\beta}\int_0^1\tau|Q(\tau x)|^4\,d\tau. \tag{2.18}$$

The possible integration constant in $(r^2e^{i\beta r^2/2}q')'=r^2e^{i\beta r^2/2}A$ vanishes because $q'=O(r^{-1})$. In Cartesian coordinates (2.17) says that $\nabla Q(x)$ is x times its continuous integral. It first gives $Q\in C^1$ and $q'=O(r)$ at zero. If Q,R are C^k locally, these compact-parameter formulas give $Q\in C^{k+1}$ and then $R\in C^{k+1}$. Differentiation is uniform on compact products. Induction proves smoothness across the origin, with uniform bounds at each fixed local derivative order. At infinity, differentiating (2.2) twice gives

$$R=O(\delta r^{-2}),\quad R'=O(\delta r^{-3}),\quad R''=O(\delta r^{-4}),\qquad \|R\|_{H^2}+\|R\|_\infty\le C\delta. \tag{2.19}$$

Choose $\delta_j\downarrow 0$ and one root θ_j for each. After a subsequence $\theta_j\to\theta_0$. Uniform contraction continuity gives convergence in X_p to $f^0_{\theta_0}$, and the limiting obstruction is zero. Let $\beta_0=\beta_*+\theta_0$ and let Q_0 be this scalar profile. The two endpoint bounds (2.14) make it nontrivial and give a nonzero coefficient at infinity.

We fix β_0 for the remainder of the proof. Rescale the positive profiles, initially denoted q_j,R_j^{old}, by

$$d_j=(\beta_0/\beta_j)^{1/2},\qquad Q_j(y)=d_jq_j(d_j|y|),\qquad R_j(y)=d_j^2R_j^{\mathrm{old}}(d_j|y|),\qquad \omega_j=d_j^2.$$

Their equations become

$$\Delta Q_j-\omega_jQ_j+i\beta_0\Lambda_1Q_j+(1+i\delta_j)|Q_j|^2Q_j-R_jQ_j=0,\qquad \beta_0\Lambda_2R_j=\delta_j|Q_j|^4. \tag{2.20}$$

For the first equation every residual term acquires d_j^3. For the second, $\beta_0\Lambda_2R_j=\beta_jd_j^4(\Lambda_2R_j^{\mathrm{old}})(d_jy)$; the fourth power on the right acquires the same factor. We have $\omega_j\to 1$ and (2.19) with $\delta=\delta_j$. Local convergence of profiles and first derivatives, followed by splitting off their uniform tails, gives

$$\begin{aligned}&\left\||Q_j|^2-|Q_0|^2\right\|_\infty+\left\||Q_j|^2-|Q_0|^2\right\|_3+\left\|\nabla(|Q_j|^2-|Q_0|^2)\right\|_3\longrightarrow 0,\\ &\left\|Q_j^2-Q_0^2\right\|_\infty+\left\|Q_j^2-Q_0^2\right\|_3+\left\|\nabla(Q_j^2-Q_0^2)\right\|_3\longrightarrow 0.\end{aligned} \tag{2.21}$$

The tail powers are r^{-2} and r^{-3}, both cubically integrable in three dimensions. Uniformly, $Q_j \in L^p$ for every $p > 3$ and $\nabla Q_j \in L^2 \cap L^3 \cap L^\infty$.

2.4 The scalar profile hypotheses for the linear estimates

Li's linear theory [3] requires a smooth nowhere-zero scalar profile, nonzero limiting modulus of rQ_0, and symbol bounds for its derivatives. We verify these properties for the particular fixed limit. Write $Q_0(x) = q(|x|)$, $\beta = \beta_0$, $\kappa = 1 + i/\beta$ and $g = r^\kappa q$. Endpoint continuity gives $g \to g_\infty \neq 0$, while the first derivative bound gives $g' = O(r^{-1})$. The scalar equation becomes

$$g'' + Dg' = -r^{-2}B(g), \qquad D = i\beta r - \frac{2i}{\beta r}, \qquad B(g) = \kappa(\kappa-1)g + |g|^2 g.$$

For $\mathfrak{m} = e^{i\beta r^2/2} r^{-2i/\beta}$ we have $\mathfrak{m}'/\mathfrak{m} = D$ and $|\mathfrak{m}| = 1$. Since $(\mathfrak{m}g')'$ is absolutely integrable and $g' \to 0$,

$$g'(r) = \mathfrak{m}(r)^{-1} \int_r^\infty \mathfrak{m}(s) s^{-2} B(g(s))\, ds. \tag{2.22}$$

For large r, $|D| \geq cr$. Put $f = r^{-2}B(g)$; initially $f = O(r^{-2})$, $f' = O(r^{-3})$. Integration by parts using $\mathfrak{m}' = D\mathfrak{m}$ gives $g' = O(r^{-3})$. The equation then gives $g'' = O(r^{-2})$, and hence $f'' = O(r^{-4})$. A second integration by parts of the remainder gives

$$g' = -f/D + O(r^{-5}), \qquad g'' = O(r^{-4}), \qquad g - g_\infty = O(r^{-2}).$$

The boundary terms vanish, since $f/D = O(r^{-3})$ and its next derivative is $O(r^{-4})$.

To obtain all derivative orders without differentiating an uncontrolled remainder, let $S^1_m(R)$ be the complete weighted C^1 space with norm

$$\|f\|_{S^1_m} = \sup_{r \geq R} (r^m |f(r)| + r^{m+1}|f'(r)|), \qquad m > 0.$$

The integral operator

$$\mathcal{V}f = -\int_r^\infty \mathfrak{m}(t)^{-1} \int_t^\infty \mathfrak{m}(s) f(s)\, ds\, dt$$

is defined with the inner oscillatory integral first. Integration by parts bounds that integral by $C_m r^{-m-1} \|f\|_{S^1_m}$; the outer integral is absolutely convergent. Thus $\mathcal{V}$ is bounded on S^1_m and $(\mathcal{V}f)'' + D(\mathcal{V}f)' = -f$. For a bounded real-linear coefficient $A(r)$ with $A' = O(r^{-1})$,

$$\left\| \mathcal{V}(r^{-2}Af) \right\|_{S^1_m} \leq C_m R^{-2} \|f\|_{S^1_m}. \tag{2.23}$$

Completeness follows by uniform convergence of weighted values and derivatives and the fundamental theorem of calculus on compact intervals.

For each integer N, choose $G_N = g_\infty + \sum_{\ell=1}^N c_\ell r^{-2\ell}$ to cancel the residual $G_N'' + DG_N' + r^{-2}B(G_N)$ through order r^{-2N}. At the new order the coefficient of c_ℓ is $-2\ell i\beta \neq 0$. The residual lies in S^1_{2N+2}. The remainder equation is

$$h = \mathcal{V}\{r^{-2}[B(G_N + h) - B(G_N)] + G_N'' + DG_N' + r^{-2}B(G_N)\}.$$

By (2.23) it is a contraction in a ball of S^1_{2N+2} for sufficiently large R. The actual remainder $g - G_N$ already lies in S^1_2. The difference v between it and the constructed remainder solves $v = \mathcal{V}(r^{-2}Av)$ on S^1_2. Here A is the real derivative of the polynomial B averaged along the segment between the

two solutions. Their bounded values and derivatives give bounded A, rA'. Enlarging R in (2.23) forces $v = 0$. Therefore $g - G_N \in S^1_{2N+2}$ for every N. Differentiation of its smooth ODE gives, for $k \geq 2$, $h^{(k)} = O(r^{-2N-4+k})$. For any specified derivative order choose N sufficiently large and differentiate $q = r^{-\kappa}(G_N + h)$. This proves

$$|q^{(k)}(r)| \leq C_k r^{-1-k} \quad (r \geq 1, k \geq 0). \tag{2.24}$$

The oscillatory homogeneous branch is excluded already by $g' \to 0$.

Finally let $n = |q|^2$ and $W = \mathrm{Im}(q'\bar{q})$. The imaginary part of the radial equation, integrated from zero, is

$$r^2 W(r) = -\frac{\beta}{2} r^3 n(r) + \frac{\beta}{2} \int_0^r s^2 n(s)\, ds.$$

A positive-radius zero would make the nonnegative integral vanish, which contradicts the nonzero origin value in (2.14). Thus q never vanishes and $\lim r|q(r)| = |g_\infty| > 0$. Together with smoothness and (2.24), this verifies conditions (1.5)–(1.8) of [3] for $d = p = 3$ and $s_c = 1/2$, including the higher-derivative condition as printed there. Here d is the spatial dimension, p is the exponent in $|u|^{p-1}u$, and $s_c = d/2 - 2/(p-1)$.

3 Estimates in similarity variables

3.1 A Green operator for one fixed scalar profile

From now on $y \in \mathbb{R}^3$ is the similarity variable. All homogeneous Sobolev spaces use the canonical realization given by Sobolev embedding; intersection norms are sums. For the single fixed profile Q_0, β_0 from Section 2, let $\delta_1 > 0$ be the spectral gap in Li's decomposition [3, Lemma 6.3]. Choose

$$\frac{1}{2} < \sigma < \min\left(0.51, \frac{1}{2} + \delta_1\right), \quad X = \dot{H}^\sigma \cap \dot{H}^1, \quad \mu = \beta_0(\sigma - 1/2), \quad 0 < \nu < \min(\mu, \beta_0/2). \tag{3.1}$$

All these choices precede the final choice of j. The real-linear scalar operator is

$$L_0 e = \Delta e - e + i\beta_0 \Lambda_1 e + 2|Q_0|^2 e + Q_0^2 \bar{e}.$$

For conjugate pairs its matrix is

$$H_0 = \begin{pmatrix} \Delta_{\beta_0} - 1 & 0 \\ 0 & -\Delta_{-\beta_0} + 1 \end{pmatrix} - \frac{i\beta_0}{2} + \begin{pmatrix} 2|Q_0|^2 & Q_0^2 \\ -\bar{Q}_0^2 & -2|Q_0|^2 \end{pmatrix}, \qquad \Delta_b = \Delta + ib(3/2 + y \cdot \nabla). \tag{3.2}$$

This is Li's operator (1.22)–(1.23) at $d = p = 3$. Its unbounded generator and flow are those in that theory. Pair spaces have component-sum norms. The involution $\mathcal{J}(z_1, z_2) = (\bar{z}_2, \bar{z}_1)$ has fixed pairs $\iota e = (e, \bar{e})$, of norm $2\,\|e\|_X$.

We state the linear facts used below. Li's Lemmas 6.3, 7.1 and 7.2 give original spectral projections P_c, P_s, P_u and compactly modified projections $\widetilde{P}_c, \widetilde{P}_s, \widetilde{P}_u$, commuting with $\mathcal{J}$, such that

$$\widetilde{P}_\alpha P_\alpha = \widetilde{P}_\alpha, \qquad P_\alpha \widetilde{P}_\alpha = P_\alpha \quad (\alpha = s, u), \tag{3.3}$$

$$\widetilde{P}_c P_c = P_c, \qquad P_c \widetilde{P}_c = \widetilde{P}_c. \tag{3.4}$$

The modified finite projections are bounded on X with finite-dimensional ranges in C_c^∞. Also $\widetilde{P}_c = I - \widetilde{P}_s - \widetilde{P}_u$ is bounded on X, and $\mathrm{Ran}\,\widetilde{P}_c = \mathrm{Ran}\,P_c$. When a projection acts on a scalar, it

means its first component after application to ιe; commutation with $\mathcal{J}$ makes this a real projection. The fixed factor two is absorbed in constants. For $z \in X \cap \operatorname{Ran} P_c$ the homogeneous estimate is

$$\left\| e^{isH_0} z \right\|_{C([k,k+1];X)} + \left\| \nabla e^{isH_0} z \right\|_{L^3([k,k+1];L^{18/5})} \le C e^{-\mu k} \|z\|_X , \qquad k \in \mathbb{Z}_{\ge 0}. \tag{3.5}$$

Indeed, (6.2) of [3] allows 18/5 when $\sigma < 0.51$; $(3, 18/5)$ is Schrödinger admissible, and the exponential weight in its Lemma 7.2 gives (3.5) on each unit interval. Time translation gives the same estimate for a later initial time.

Define the solution and forcing norms

$$\|e\|_{\mathcal{S}_\nu} = \sup_{k\ge 0} e^{\nu k} \left(\|e\|_{C([k,k+1];X)} + \|\nabla e\|_{L^3([k,k+1];L^{18/5})} \right), \tag{3.6}$$

$$\|F\|_{\mathcal{F}_\nu} = \sup_{k\ge 0} e^{\nu k} \|F\|_{L^1([k,k+1];X)} . \tag{3.7}$$

The central-stable initial space is $(I - \widetilde{P}_u)X$.

Lemma 3.1. *There is a bounded real-linear solution operator $\mathcal{B}$ for $ie_s + L_0 e = F$, with prescribed central-stable initial component $e_{cs,0} \in (I - \widetilde{P}_u)X$, such that*

$$\|\mathcal{B}(e_{cs,0}, F)\|_{\mathcal{S}_\nu} \le C_B(\|e_{cs,0}\|_X + \|F\|_{\mathcal{F}_\nu}). \tag{3.8}$$

Its remaining initial component lies in $\operatorname{Ran} \widetilde{P}_u \subset C_c^\infty$. *All constants and projections depend on $Q_0, \beta_0, \sigma, \nu$, not on j.*

Proof. Choose real bases $(\widetilde{\varphi}_{\alpha,\ell})_\ell$ of the $\mathcal{J}$-fixed parts of $\operatorname{Ran} \widetilde{P}_\alpha$ for $\alpha = s, u$. The finite coordinate vectors z_α reconstruct the pair-valued modes

$$\widetilde{Z}_\alpha(s) = \sum_\ell (z_\alpha(s))_\ell \widetilde{\varphi}_{\alpha,\ell}, \qquad \alpha = s, u.$$

The central component is $Z_c = \widetilde{P}_c \iota e$. In Li's modified coordinate equations (7.14), the finite coordinates obey $z'_\alpha + A_\alpha z_\alpha = f_\alpha$. Here A_α realizes $-i\widetilde{P}_\alpha H_0 P_\alpha$ in real finite coordinates, and f_α realizes $-i\widetilde{P}_\alpha(F, -\bar{F})$. The stable eigenvalues have real part at least $\beta_0\delta_1 > \mu$, whereas the unstable eigenvalues have real part at most zero. Use the forward solution for the stable equation and

$$z_u(s) = -\int_s^\infty e^{A_u(\tau - s)} f_u(\tau)\, d\tau$$

for the unstable one. On a Jordan block the exponential is an exponential times a polynomial. The stable forward kernel is bounded by $Ce^{-\gamma_s d}$ for some $\gamma_s > \mu$; the backward unstable kernel grows at most polynomially. Their convolutions with $e^{-\nu k}$ obey the required weighted bounds.

The central forcing is $\widetilde{P}_c(F, -\bar{F})$ together with $-\widetilde{P}_c H_0(\widetilde{Z}_s + \widetilde{Z}_u)$. Every finite modified basis vector is compact and smooth, so its image under H_0 and then $\widetilde{P}_c$ belongs to X. Thus this additional forcing is controlled by the finite coordinates. Apply (3.5) to each forcing impulse, use Minkowski's inequality, and sum

$$\sum_{\ell=0}^{k} e^{-\mu(k-\ell-1)_+} e^{-\nu\ell} \le C_{\mu,\nu} e^{-\nu k}. \tag{3.9}$$

An impulse inside the observation interval uses the unweighted homogeneous estimate on the remaining subinterval. Its norm is bounded by that interval's L^1X forcing norm. The finite-coordinate sums are treated with the same geometric sum, or a convergent polynomial-exponential sum in the backward direction. This proves (3.8).

The scalar real structure and signs are consistent: direct calculation gives $\mathcal{J}H_0\mathcal{J} = -H_0$, so e^{isH_0} commutes with $\mathcal{J}$. The pair forcing $(F, -\bar{F})$ is anti-invariant, and its generator forcing $-i(F, -\bar{F})$ is invariant. The modified projections preserve this fixed real subspace. The construction therefore returns an actual scalar e and its conjugate. □

The forcing norm in (3.7) permits a small constant multiple of e, since each interval has length one. This will accommodate the frequency change $\omega_j - 1$ without requiring an L^2 bound for the similarity perturbation.

3.2 Products at the quartic density nonlinearity

For $I = [k, k+1]$ abbreviate

$$a_I(e) = \|e\|_{C(I;X)} + \|\nabla e\|_{L^3(I;L^{18/5})}.$$

Sobolev embedding and interpolation give $X \hookrightarrow L^4 \cap L^6$. We also need an estimate combining the derivative Strichartz norm and L^6. Averaging a smooth f on a radius-R ball and integrating its gradient along segments gives

$$|f(x)| \le CR^{-1/2}\|f\|_6 + C\int_{|x-z|<R} |x-z|^{-2}|\nabla f(z)|\,dz \le C(R^{-1/2}\|f\|_6 + R^{1/6}\|\nabla f\|_{18/5}).$$

The kernel belongs locally to $L^{18/13}$, since $36/13 < 3$. Balancing $R = (\|f\|_6 / \|\nabla f\|_{18/5})^{3/2}$ gives

$$\|f\|_\infty \le C\|\nabla f\|_{18/5}^{3/4}\|f\|_6^{1/4}. \tag{3.10}$$

The zero cases follow by a limit. Mollification and an almost-everywhere subsequential limit in L^6 extend the estimate to its stated norm class. Interpolation also gives $\|\nabla e\|_3 \le \|\nabla e\|_2^{1/4}\|\nabla e\|_{18/5}^{3/4}$. Raising to the fourth power and integrating in time yields

$$\|e\|_{L^4(I;L^\infty)} + \|\nabla e\|_{L^4(I;L^3)} + \|e\|_{C(I;L^4\cap L^6)} \le Ca_I(e). \tag{3.11}$$

Recall from the end of Section 2.3 that Q_j is uniformly bounded, belongs uniformly to L^p for each fixed $p > 3$, and has gradient uniformly in $L^2 \cap L^3 \cap L^\infty$. Set

$$S_j(e) = |Q_j + e|^4 - |Q_j|^4, \qquad N_j(e) = (1 + i\delta_j)(2Q_j|e|^2 + \bar{Q}_j e^2 + |e|^2 e).$$

Writing $a = a_I(e)$ and $b = \|h\|_{C(I;H^1)}$, we have

$$\|S_j(e)\|_{L^1(I;H^1)} \le C(a + a^2 + a^3 + a^4), \tag{3.12}$$

$$\|N_j(e)\|_{L^1(I;H^1)} \le C(a^2 + a^3), \tag{3.13}$$

$$\|h(Q_j + e)\|_{L^1(I;H^1)} \le Cb(1 + a). \tag{3.14}$$

Here and below these constants are uniform for all sufficiently large j. For clarity, expand $|Q+e|^2 = |Q|^2 + 2\,\mathrm{Re}(Q\bar{e}) + |e|^2$ and square. The linear term of S_j is in L^2 by $Q_j^3 \in L^3$ and $e \in L^6$. Its derivative uses $Q_j^3\nabla e$ and $Q_j^2\nabla Q_j\, e$. The quadratic undifferentiated terms use $e \in L^4$, and their derivatives use $\|e\|_\infty\|\nabla e\|_2$. The cubic terms use $\|e\|_6^3$ and $\|e\|_\infty^2\|\nabla e\|_2$. At degree four,

$$\left\|e^4\right\|_2 \le \|e\|_6^3\|e\|_\infty, \qquad \left\|\nabla(e^4)\right\|_2 \le 4\|e\|_\infty^3\|\nabla e\|_2.$$

Derivatives on profile factors are controlled by their uniform bounds. Time Hölder and (3.11) prove (3.12); the quadratic and cubic terms also prove (3.13). For the density product use the bounded profile and

$$\|he\|_2 \le \|h\|_3 \|e\|_6 , \quad \|h\nabla e\|_2 \le \|h\|_6 \|\nabla e\|_3 , \quad \|e\nabla h\|_2 \le \|e\|_\infty \|\nabla h\|_2 .$$

These H^1 forcings belong to X, because $0 < \sigma < 1$.

The estimates also hold for differences. Factoring a difference of products as a sum with one difference factor, before and after one derivative, gives

$$\|S_j(e_1) - S_j(e_2)\|_{L^1H^1} \le C(1 + a_1 + a_2)^3 a_I(e_1 - e_2), \tag{3.15}$$

$$\|N_j(e_1) - N_j(e_2)\|_{L^1X} \le C(a_1 + a_2 + (a_1 + a_2)^2) a_I(e_1 - e_2), \tag{3.16}$$

$$\|h_1(Q_j + e_1) - h_2(Q_j + e_2)\|_{L^1X} \le C[\|h_1 - h_2\|_{CH^1} (1 + a_1) + \|h_2\|_{CH^1} a_I(e_1 - e_2)]. \tag{3.17}$$

The omitted time interval in these three norms is I, and $a_i = a_I(e_i)$. Finally, for a coefficient V,

$$\|Ve\|_{H^1} \le C(\|V\|_3 + \|V\|_\infty + \|\nabla V\|_3) \|e\|_X . \tag{3.18}$$

Equations (2.21), (2.19) and (3.18) make the profile-potential changes small from X to H^1. The constant multiplier $\omega_j - 1$ is instead small on X itself. This distinction is useful because Q_j and the initial cutoff correction need not be in L^2 separately.

3.3 The density propagator in the optical frame

Let $e_z = (0, 0, 1)$ and introduce the optical frame $X_{\rm ph} = x - te_z$, $\widehat{u}(t, X_{\rm ph}) = u(t, X_{\rm ph} + te_z)$, with the same convention for $\widehat{\rho}$. The specialized original equations become

$$i\widehat{u}_t + \Delta\widehat{u} + (1 + i\delta_j)|\widehat{u}|^2\widehat{u} - \widehat{\rho}\widehat{u} = 0, \qquad \widehat{\rho}_t = \delta_j|\widehat{u}|^4 + \partial_z\widehat{\rho}.$$

For $0 < \lambda_0 \le 1$ define

$$\lambda(s) = \lambda_0 e^{-\beta_0 s}, \quad t(s) = \frac{\lambda_0^2}{2\beta_0}(1 - e^{-2\beta_0 s}), \quad y = X_{\rm ph}/\lambda(s), \quad T = \frac{\lambda_0^2}{2\beta_0}. \tag{3.19}$$

Since $dt/ds = \lambda^2$, the ansatz

$$\widehat{u} = \lambda^{-1} e^{i\omega_j s}(Q_j + e), \qquad \widehat{\rho} = \lambda^{-2}(R_j + h) \tag{3.20}$$

and (2.20) give the exact density equation

$$h_s + \beta_0\Lambda_2 h - \lambda(s)\partial_z h = \delta_j S_j(e) + \lambda(s)\partial_z R_j, \qquad h(0) = -R_j. \tag{3.21}$$

Thus the optical translation does not disappear from the density equation. On compact intervals before T, the coordinate change is a smooth diffeomorphism with nonzero Jacobian; the calculation also holds distributionally by pulling back test functions.

Solving the characteristic equation $y' = \beta_0 y - \lambda(s)e_z$ gives, for $s \ge \tau$ and $d = s - \tau$,

$$(\mathcal{T}(s, \tau)f)(y) = e^{-2\beta_0 d} f\left(e^{-\beta_0 d} y + \frac{\lambda(\tau)}{2\beta_0}(1 - e^{-2\beta_0 d})e_z\right). \tag{3.22}$$

Change of variables gives

$$\|\mathcal{T}f\|_2 = e^{-\beta_0 d/2} \|f\|_2 , \qquad \|\nabla\mathcal{T}f\|_2 = e^{-3\beta_0 d/2} \|\nabla f\|_2 . \tag{3.23}$$

The propagator is strongly continuous on H^1, first on Schwartz functions and then by density and these norm bounds. Let $\mathcal{Y}_\nu$ be the real space $C([0,\infty);H^1)$ with norm $\|h\|_{\mathcal{Y}_\nu} = \sup_{s\geq 0} e^{\nu s}\|h(s)\|_{H^1}$. Denote the integral solution of (3.21) by $\Phi_\rho(e)$. Equations (3.12), (3.23) and $\nu < \beta_0/2$ imply

$$\|\Phi_\rho(e)\|_{\mathcal{Y}_\nu} \leq C\delta_j(1+\lambda_0+E+E^2+E^3+E^4), \tag{3.24}$$

$$\|\Phi_\rho(e_1)-\Phi_\rho(e_2)\|_{\mathcal{Y}_\nu} \leq C\delta_j(1+E_1+E_2)^3\|e_1-e_2\|_{\mathcal{S}_\nu}, \tag{3.25}$$

where $E = \|e\|_{\mathcal{S}_\nu}$ and $E_i = \|e_i\|_{\mathcal{S}_\nu}$. For example the background forcing is bounded in H^1 by $C\delta_j\lambda_0 e^{-\beta_0 s}$, since $\|\partial_z R_j\|_{H^1} \leq C\delta_j$. The other terms are summed interval by interval using (3.9) with density decay rate $\beta_0/2$ in place of μ. The map preserves real-valued h.

4 A singular solution from compact smooth data

4.1 Solving the coupled perturbation equations

With the profiles and spaces now fixed, put

$$L_j e = \Delta e - \omega_j e + i\beta_0\Lambda_1 e + (2|Q_j|^2 - R_j)e + Q_j^2\bar{e} + i\delta_j(2|Q_j|^2 e + Q_j^2\bar{e}).$$

Subtracting (2.20) from the optical equation in (3.20) gives

$$ie_s + L_0 e = -(L_j - L_0)e - N_j(e) + h(Q_j+e) =: F_j(e,h). \tag{4.1}$$

For a prescribed small $e_{cs,0} \in (I-\widetilde{P}_u)X$, define $\Phi_u(e,h) = \mathcal{B}(e_{cs,0}, F_j(e,h))$. We use the product norm

$$\|(e,h)\|_{\mathcal{Z}_j} = \|e\|_{\mathcal{S}_\nu} + \delta_j^{-1/2}\|h\|_{\mathcal{Y}_\nu}.$$

These are complete spaces. In particular a locally uniform X limit and an $L^3L^{18/5}$ limit of its gradients agree as distributions; the conjugate-paired real optical subspace is closed. The density weight makes both directions of the feedback small. Writing $E = \|e\|_{\mathcal{S}_\nu}$ and $H = \delta_j^{-1/2}\|h\|_{\mathcal{Y}_\nu}$, the preceding estimates give some $\eta_j \to 0$ such that

$$\|\Phi_u(e,h)\|_{\mathcal{S}_\nu} \leq C[\|e_{cs,0}\|_X + \eta_j E + E^2 + E^3 + \sqrt{\delta_j}H(1+E)], \tag{4.2}$$

$$\delta_j^{-1/2}\|\Phi_\rho(e)\|_{\mathcal{Y}_\nu} \leq C\sqrt{\delta_j}(1+\lambda_0+E+E^2+E^3+E^4). \tag{4.3}$$

On interval $[k,k+1]$, $a_I(e) \leq Ee^{-\nu k}$ and $\|h\|_{CH^1} \leq \sqrt{\delta_j}He^{-\nu k}$. Multiplying the forcing estimate by $e^{\nu k}$ gives the displayed polynomials, including the linear feedback term. The frequency part $(\omega_j - 1)e$ is estimated directly in X.

On a ball of radius $r < 1$ in $\mathcal{Z}_j$, the difference estimates give Lipschitz constant at most

$$C[\eta_j + r + r^2 + \sqrt{\delta_j}(1+r)^3].$$

First choose r small and then j sufficiently large to make this less than $1/2$. Fix a sufficiently small $\lambda_0 > 0$ and choose $\|e_{cs,0}\|_X$ small. Increasing j if needed makes the constant terms in (4.2)–(4.3) sum to at most $r/2$. The map (Φ_u, Φ_ρ) is therefore a contraction of the closed ball. It has a fixed point satisfying both exact equations, with

$$\|e(s)\|_X + \|h(s)\|_{H^1} \longrightarrow 0 \qquad (s\to\infty). \tag{4.4}$$

The spectral constants and projections in this argument belong to the one fixed scalar profile. Only the bounded coefficient differences in (2.21) are perturbed.

4.2 Choosing compact initial data

Let $\chi \in C_c^\infty(\mathbb{R}^3)$ equal one on a ball, and $\chi_L(y) = \chi(y/L)$. For the selected j, the tail $w_L = (\chi_L - 1)Q_j$ tends to zero in X. To verify this in the homogeneous norm, partition the tail into dyadic annuli of radii $R \geq L$. Each annular term f_R obeys

$$\|f_R\|_2 \leq CR^{1/2}, \qquad \|\nabla f_R\|_2 \leq CR^{-1/2}, \qquad \|f_R\|_{\dot{H}^\sigma} \leq \|f_R\|_2^{1-\sigma} \|\nabla f_R\|_2^{\sigma} \leq CR^{1/2-\sigma}.$$

The two geometric series converge and tend to zero as $L \to \infty$, because $\sigma > 1/2$. Local finiteness identifies their sum with the actual tail as a distribution.

Choose the central-stable datum $e_{cs,0} = (I - \widetilde{P}_u)w_L$. It is small for large L. Lemma 3.1 supplies an initial correction $v_u \in \operatorname{Ran} \widetilde{P}_u \subset C_c^\infty$, so

$$Q_j + e(0) = \chi_L Q_j + v_u - \widetilde{P}_u w_L \in C_c^\infty. \tag{4.5}$$

The resulting physical data are

$$u_0(x) = \lambda_0^{-1}[Q_j + e(0)](x/\lambda_0), \qquad \rho_0(x) = \lambda_0^{-2}[R_j + h(0)](x/\lambda_0) = 0. \tag{4.6}$$

The optical datum is compact and smooth, despite the nonintegrable mass tail of the profile before cancellation.

4.3 Identification with the physical Cauchy solution

For completeness we give the local theory used to identify the constructed solution. The physical equations at the selected parameters are

$$i(\partial_t + \partial_z)u + \Delta u = -(1 + i\delta_j)|u|^2 u + \rho u, \qquad \rho_t = \delta_j |u|^4. \tag{4.7}$$

The free optical group is $S(t)f(x) = (e^{it\Delta} f)(x - te_z)$. Its spatial norms equal those of the free Schrödinger group. The free retarded Strichartz estimates [4, Theorem 2.3], with derivatives, control

$$A_I(u) = \|u\|_{C(I;H^1)} + \|\nabla u\|_{L^3(I;L^{18/5})}$$

from H^1 data and forcings in $L^{4/3}W^{1,3/2} + L^2 W^{1,6/5}$. The pairs $(4,3)$ and $(2,6)$ are admissible in dimension three. The source normalization $\Delta/2$ changes to Δ by $t \mapsto 2t$, which changes the time norms only by fixed factors. Estimate (3.10) gives $\|u\|_{L^4(I;L^\infty)} \leq C A_I(u)$.

For $|I| \leq 1$, Hölder and the product rule give

$$\left\| |u|^4 \right\|_{L^1 H^1} \leq |I|^{3/4} \|u\|_{CL^6}^3 \|u\|_{L^4 L^\infty} + 4|I|^{1/4} \|u\|_{L^4 L^\infty}^3 \|\nabla u\|_{CL^2}, \tag{4.8}$$

$$\|\rho u\|_{L^{4/3}W^{1,3/2}} \leq C|I|^{3/4} \|\rho\|_{CH^1} \|u\|_{CH^1}, \tag{4.9}$$

$$\left\| |u|^2 u \right\|_{L^2 W^{1,6/5}} \leq C|I|^{1/2} \|u\|_{CH^1}^3. \tag{4.10}$$

All time norms here are on I. For the derivative in (4.9), use $\nabla\rho \in L^2, u \in L^6$ and $\rho \in L^6, \nabla u \in L^2$; the undifferentiated product uses $\rho \in L^2, u \in L^6$. For the cubic derivative use two L^6 factors and one L^2 derivative; for the undifferentiated term use two L^6 factors and one L^2 factor. The quartic bounds are the same degree-four estimates as above.

Differences are estimated by factoring products. In particular, a differentiated quartic difference is bounded by terms of the form

$$(|u| + |v|)^3 |\nabla(u - v)| \quad \text{and} \quad (|u| + |v|)^2 |u - v|(|\nabla u| + |\nabla v|).$$

Three $L^4_t L^\infty_x$ factors and one $C_t L^2_x$ derivative give $|I|^{1/4}$. The undifferentiated difference uses three $C_t L^6_x$ factors and one $L^4_t L^\infty_x$ difference, giving $|I|^{3/4}$. Thus the optical Duhamel map and $\rho = \rho_0 + \delta_j \int |u|^4$ contract on a sufficiently short interval in the class $A_I \times C(I; H^1)$, for each fixed H^1 datum. This proves local existence and uniqueness in that class.

The reconstruction (3.20) initially gives $u \in C_{\mathrm{loc}}([0,T); X)$, $\rho \in C_{\mathrm{loc}}([0,T); H^1)$ and the local gradient Strichartz bound. On a compact physical interval the similarity time interval is compact and all scale factors are bounded above and below. In particular

$$\left\| \lambda^{-1} f(\cdot/\lambda) \right\|_{\dot H^a} = \lambda^{1/2-a} \, \|f\|_{\dot H^a} \quad (a = \sigma, 1),$$

whereas the density L^2 and $\dot H^1$ factors are $\lambda^{-1/2}$ and $\lambda^{-3/2}$. Translations, phases and dilations are strongly continuous on these spaces, as follows first on Schwartz functions and then by completion. The background gradient belongs to $L^{18/5}$ by its uniform decay. The exact coordinate calculation proves (4.7) distributionally for this reconstruction.

To recover the missing mass norm, observe that $u \in CL^6$ and $\rho \in CL^3$ on compact intervals, so the optical forcing belongs to CL^2:

$$\left\| |u|^2 u \right\|_2 = \|u\|_6^3, \qquad \|\rho u\|_2 \le \|\rho\|_3 \, \|u\|_6 .$$

Since (4.6) belongs to L^2, its physical Duhamel formula is in CL^2. It equals the reconstructed distributional solution: their difference solves the free equation with zero forcing and zero initial trace; spatial Fourier transformation and the Schrödinger integrating factor, together with the transport translation, make it zero. Consequently $u \in CH^1$ on every compact interval before T, and the local uniqueness just proved applies.

The original density is therefore

$$\rho(t,x) = \delta_j \int_0^t |u(\tau,x)|^4 \, d\tau \ge 0 \tag{4.11}$$

almost everywhere. To identify the H^1 integral with this scalar integral, use (4.8) and spatial Hölder on bounded boxes: the absolute spacetime integral is finite there. Fubini–Tonelli then applies. Positivity concerns the actual density even though h can have either sign.

4.4 Smooth persistence and divergence

For each integer $m \ge 3$, H^m embeds into L^∞ and is a product algebra. The physical Duhamel and density integral maps are locally Lipschitz on $C(I; H^m \times H^m)$, with a lifespan bounded below in terms of the initial norms. These solutions agree with the preceding H^1 solution by uniqueness. On every $[0,T']$ with $T' < T$, the construction gives $u \in L^4_t L^\infty_x$. Equation (4.11) implies

$$\sup_{t \le T'} \|\rho(t)\|_\infty \le \delta_j \int_0^{T'} \|u(t)\|_\infty^4 \, dt < \infty.$$

The tame Sobolev product estimate [4, Appendix A, Lemma A.8] applied to (4.7) yields, for $Y_m = \|u\|_{H^m} + \|\rho\|_{H^m}$,

$$Y_m(t) \le Y_m(0) + C_{m,\delta_j} \int_0^t [1 + \|u\|_\infty^2 + \|u\|_\infty^3 + \|\rho\|_\infty] Y_m(\tau) \, d\tau. \tag{4.12}$$

The optical potential contributes $\|\rho\|_\infty \|u\|_{H^m} + \|u\|_\infty \|\rho\|_{H^m}$, the cubic contributes $C \|u\|_\infty^2 \|u\|_{H^m}$, and the quartic density source contributes $C \|u\|_\infty^3 \|u\|_{H^m}$. The bound follows directly from the

Duhamel norm inequality and does not differentiate a norm through the unbounded Laplacian. Its coefficient is integrable on $[0, T']$. The integral Gronwall inequality gives a uniform H^m bound there. If a smooth lifespan ended at $S < T$, the constructed solution would supply the same integrable coefficient beyond S. The uniform norm bound and the norm-dependent local lifespan would extend it, a contradiction. Thus every fixed m persists on $[0, T)$; uniqueness identifies all these solutions. Density nonnegativity is now pointwise.

Proof of Theorem 1.1. Select the finite j, the scale λ_0 and the cutoff L as above. The data (4.6) are Schwartz, with $\rho_0 = 0$. We have obtained the original smooth solution on $[0, T)$, with all the claimed Sobolev regularity and nonnegative density. The norm $\|\nabla Q_j\|_2$ is positive: a zero gradient would make Q_j constant, contrary to its nonzero value and decay. By (4.4),

$$\|\nabla(Q_j + e(s))\|_2 \longrightarrow \|\nabla Q_j\|_2 > 0.$$

Undoing the dilation, phase and translation gives

$$\|\nabla u(t(s))\|_2 = \lambda(s)^{-1/2} \|\nabla(Q_j + e(s))\|_2 \longrightarrow \infty, \qquad t(s) \uparrow T < \infty. \tag{4.13}$$

The datum u_0 cannot vanish: uniqueness would then give the zero optical solution, contradicting (4.4) and $Q_j \neq 0$. The chosen coefficient is the strictly positive δ_j, rather than its zero limit. Also $\epsilon = c = c_g = 1$, $K = 2$, and $\alpha_5 = 0$ are admissible in the original system. The Laplacian has all three elliptic spatial directions, and the density translation was retained exactly. A global $C_t(H^3 \times H^3)$ solution would coincide with this one before T and have bounded gradient norm near T, contradicting (4.13). This proves the asserted counterexample. □

A Finite exact coefficient inequalities

Only a small additional arithmetic calculation is needed beyond the quantitative scalar theorems in [2]. Recall the constants from Section 2.1:

$$\beta_* = \frac{772201763088846}{841768781900003}, \qquad r_* = 1.2 \cdot 10^{-6}, \qquad \beta_{\min} = \beta_* - 10^{-10}, \qquad \beta \in [\beta_{\min}, \beta_* + 10^{-10}].$$

For its published polynomial $P_* = \sum_{n=0}^{50} c_n T_n$, set

$$U = \sum c_n, \qquad V = \sum (-1)^n c_n, \qquad W = \sum n^2 c_n, \qquad |z|_1 = |\operatorname{Re} z| + |\operatorname{Im} z|, \qquad \phi(\beta) = \frac{2i\beta}{1 - i\beta}.$$

The identities $T_n(1) = 1$, $T_n(-1) = (-1)^n$ and $T_n'(1) = n^2$ follow from the Chebyshev recurrence and its derivative. Also $|T_n(y)| \leq 1$ on $[-1, 1]$, and $|\phi| \leq 2$, $|\phi'| \leq 2$ for positive real β. Consequently (2.14) follows from the three exact rational comparisons

$$2 \sum_{n=0}^{50} |c_n|_1 + 2 \sum_{n=0}^{50} n^2 |c_n|_1 + r_* < 50, \tag{A.1}$$

$$\operatorname{Re}[V - U + \phi(\beta_*) W] + 2 \cdot 10^{-10} |W|_1 + r_* < -1, \tag{A.2}$$

$$\frac{4\beta_{\min}^2}{1 + \beta_{\min}^2} |W|^2 > \frac{1}{100}. \tag{A.3}$$

The last bound uses that $\beta^2/(1 + \beta^2)$ increases for $\beta > 0$. In particular $|F(1)| > 1/10 - r_* > 0$.

The companion package supplies the 51 complex rational coefficients from Appendix A of the pinned version of [2], and the standalone script `code/check_constants.py`. Its arithmetic uses Python's exact `Fraction` type. It checks (A.1)–(A.3), the self-map inequality, (2.15), the two obstruction margins and the positive-δ bounds in (2.12)–(2.13). The endpoint sums are checked independently by constructing the Chebyshev polynomials in the monomial basis. Input hashes and a replay command are in the supplement's `code/README.md`. No floating-point comparison, spatial discretization or PDE simulation is involved. This finite calculation does not reproduce the full computer-assisted inverse and residual proof of [2]; those published theorems remain external dependencies. The bound 10^{-23} used for the profile perturbation is not a numerical threshold for the full Cauchy construction, which also requires the qualitative choices after (4.3).

AI use

AI assistance was used for mathematical exploration, preparation of the proof and manuscript, exact-arithmetic code, and internal proof checking. Two further AI reviews in fresh contexts, a reader simulation and a referee simulation, examined the same anonymized draft. They shared the working environment. These activities do not constitute independent human verification or formal proof certification. The argument uses the published scalar computer-assisted construction of Donninger and Schörkhuber and the linear estimates of Li as cited above.

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